Osmosis and Diffusion

Water moves across a selectively permeable membrane from a region of lower solute concentration to a region of higher solute concentration. This net movement of water is called osmosis. When the solute concentration is equal on both sides the cell is isotonic.

Cell Wall and Membrane

The plasma membrane is a phospholipid bilayer that controls what enters and leaves the cell. The cell wall is a rigid structure outside the membrane that provides support and shape to plant cells.

Photosynthesis Overview

Photosynthesis converts light energy into chemical energy stored in glucose. It happens in the chloroplasts of plant cells and involves the light dependent reactions followed by the Calvin cycle.

Enzyme Catalysis

Enzymes are biological catalysts that lower the activation energy of a reaction. Each enzyme has an active site that binds a specific substrate, and the reaction rate depends on temperature and pH.